

A Distributed Approach for Mitigating Beam-Split Effects in IRS-Aided Wideband Systems

Group meeting - Main presentation

Yashvanth L.

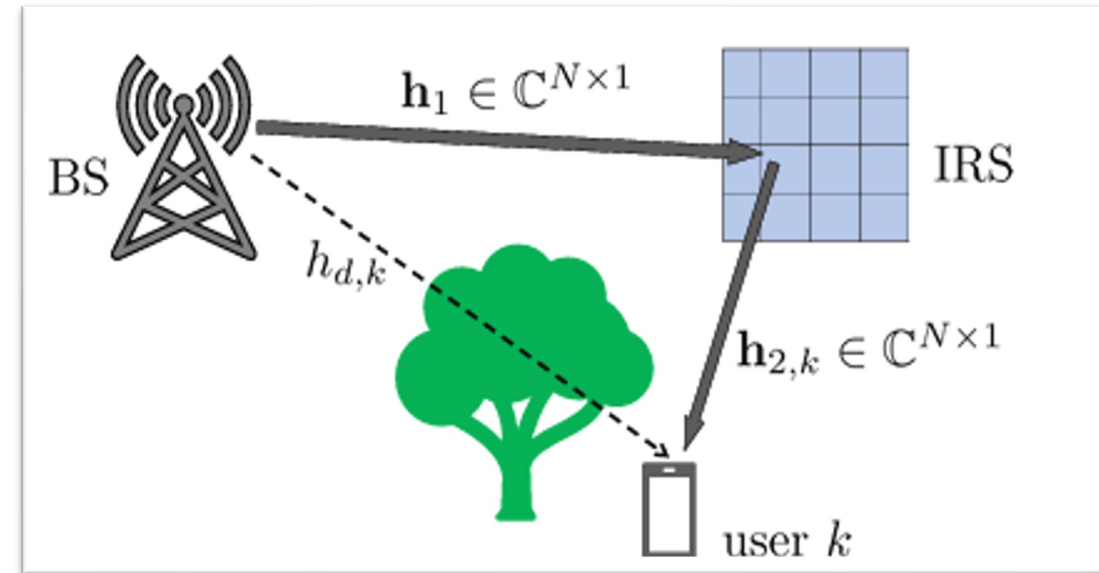
Dept. of Electrical Communication Engineering,
Indian Institute of Science, Bangalore

Email: yashvanthl@iisc.ac.in

February 17, 2025

Intelligent Reflecting Surface (IRS)

- IRSs: Also known as reconfigurable intelligent surfaces (RIS)
- Meta surfaces with passive elements
 - ❑ Alters the overall channel - *"reflect the signals in required directions"*
- Perks
 - ❑ Boosts SNR/SINR
 - ❑ Energy efficiency
 - ❑ Coverage enhancement, etc.
- State-of-the-art controlling mechanism:
 - ❑ CSI estimation
 - ❑ IRS phase optimization
 - ❑ Transportation of phase angles from BS to IRS



$$h_{k,q}(t) = \sqrt{\beta_{r,k}} \mathbf{h}_{2,k}^H \mathbf{\Theta}_q(t) \mathbf{h}_1 + \sqrt{\beta_{d,k}} h_{d,k}.$$

TODAY: IRS-aided wideband beamforming - Challenges & Solutions

Beamforming in mm-Wave Bands

- ❑ FR-2 Bands in the 5G standards (24.25 GHz - 52.6 GHz, Rel. 15, 2018)
- ❑ Channels are highly **directional**!
- ❑ **Sparse** channels with **few** propagation **paths**
 - ❑ Mostly **LoS** propagation



System Model in Wideband mmWave Scenarios

- Setting: N -element IRS in a **point-point mmWave** system with **LoS** propagation

- Baseband channel from BS to n^{th} IRS element with **DoA** ψ :

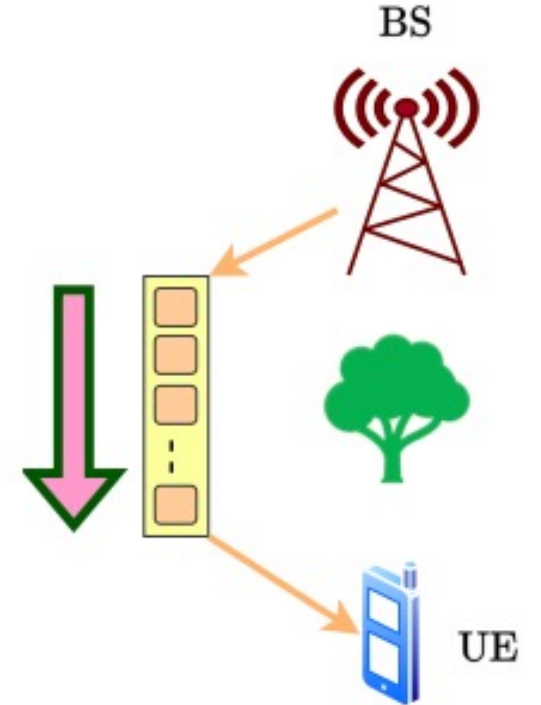
$$h_{1,n}(t) = \alpha \delta \left(t - \eta^{(1)} - (n-1) \frac{d}{c} \sin(\psi) \right) e^{-j2\pi f_c (n-1) \frac{d}{c} \sin(\psi)}$$

- Baseband channel from n^{th} IRS element to UE with **DoD** ω :

$$h_{2,n}(t) = \gamma \delta \left(t - \eta^{(2)} + (n-1) \frac{d}{c} \sin(\omega) \right) e^{-j2\pi f_c (n-1) \frac{d}{c} \sin(\omega)}$$

- Baseband channel from BS to UE with **IRS configuration** $\{\theta_n\}$:

$$\begin{aligned} h(t) &= \sum_{n=1}^N \theta_n h_{2,n}(t) \circledast h_{1,n}(t) \\ &= \sum_{n=1}^N \theta_n \alpha \gamma \delta \left(t - \eta - (n-1) \frac{d}{c} (\sin(\psi) - \sin(\omega)) \right) \times e^{-j2\pi f_c (n-1) \frac{d}{c} \sin(\phi)} \end{aligned}$$



Narrowband condition fails in large IRS aided mmWaves

- The overall channel:

$$h(t) = \sum_{n=1}^N \theta_n \alpha \gamma \delta \left(t - \eta - (n-1) \frac{d}{c} (\sin(\psi) - \sin(\omega)) \right) e^{-j2\pi f_c (n-1) \frac{d}{c} \sin(\phi)}$$

- The bulk delay, η , can be compensated using a **timing offset** @ the receiver
- Delay spread: $\Delta\tau^C = (N-1) \frac{d}{c} (\sin(\psi) - \sin(\omega))$
- Narrowband condition: "*Delay spread $\ll$ sampling time*": $\Delta\tau^C \ll T_s = 1/W$
- **Some numbers with large IRSs**: $N = 1024$, $f_c = 30$ GHz, $W = 400$ MHz, $\sin(\psi) - \sin(\omega) = 1$:
Delay spread: $\Delta\tau^C \approx 1.7 \times 10^{-8}$, sampling time: $T_s = 2.5 \times 10^{-9}$ **Narrowband condition fails!**
- Large arrays cause *spatial delay spread*, giving rise to the *spatial wideband effect (SWE)*

The curse of Spatial wideband effect: The Beam-split

- Spatial wideband effect in **time domain** $\Rightarrow$ beam-split effect in **frequency domain**

- The frequency domain channel:

$$H(f) = \beta \sum_{n=1}^N \theta_n e^{-j\pi(n-1) \sin(\phi)} \left\{ 1 + \frac{f}{f_c} \right\} = \sqrt{N} \beta \boldsymbol{\theta}^H \mathbf{a}_N \left(\sin_{(p)}^{-1} \left\{ (1 + (f/f_c)) \sin(\phi) \right\} \right)$$

Frequency-dependent direction!

- Frequency-independent IRS phase shifters: **cannot beamform** to UE over **full BW**

- Say, IRS aligns to UE at $f = 0$: $\boldsymbol{\theta} = \mathbf{a}_N(\phi)$

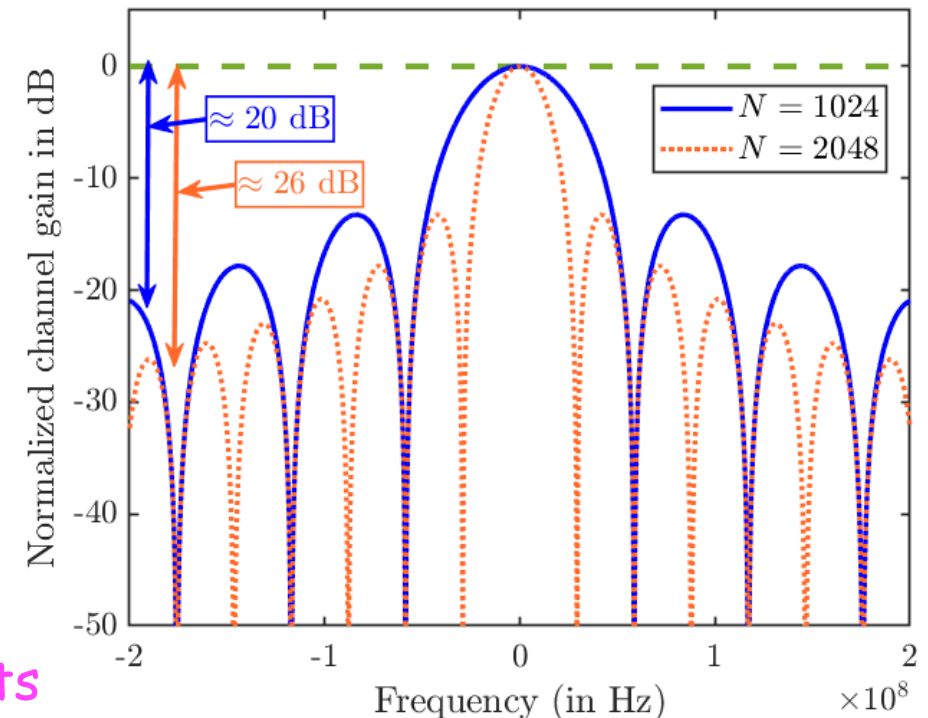
- ☐ Channel response on sub-carrier k :

$$|H_C[k]|^2 = N^2 |\beta|^2 \text{sinc}^2 \left(N \frac{f_k}{2f_c} \sin(\phi) \right)$$

✓ From the above e.g., $N \leq 128$ to remain within HPBW

- ☐ Array **gain degrades** on SCs with $f \neq 0$ - the **beam split**

- ☐ **Limits** the allowed **bandwidth** or **number of IRS elements**



Problem: Handle Beam Split in IRS-aided wideband systems

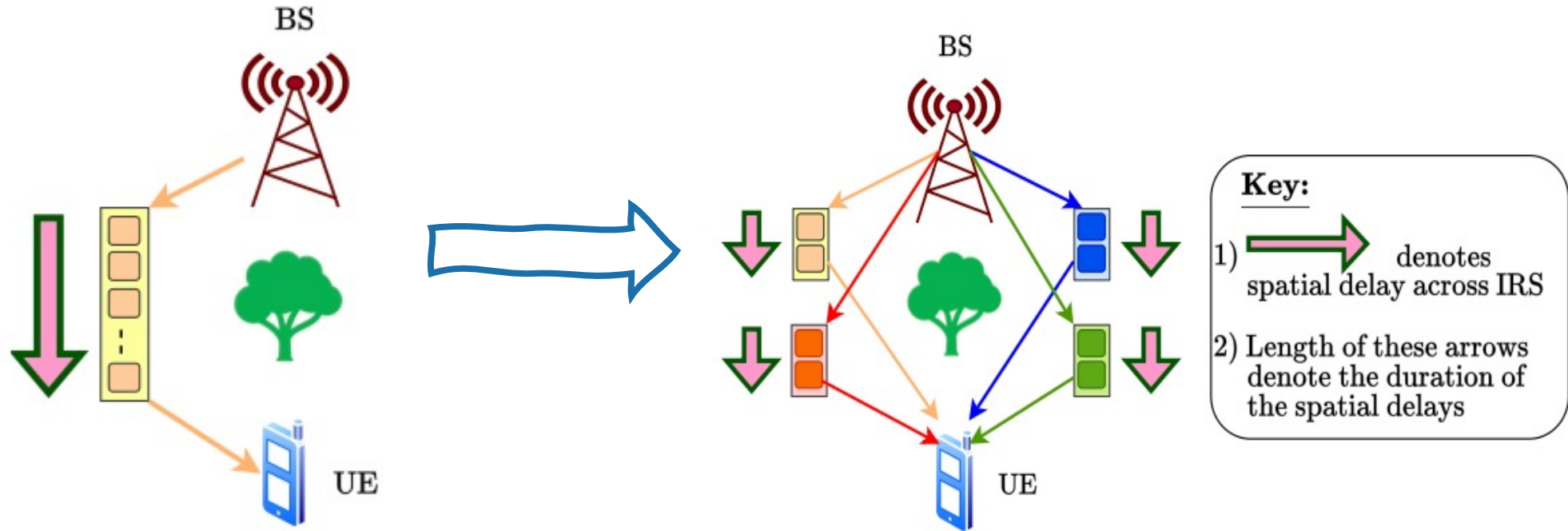
- **State-of-the-art:** Use **true-time-delay (TTD)** units at the IRS**
 - ❑ IRS element response: $\theta_n(t) = e^{j\phi_n} \delta(t - \tau_n)$
 - ❑ **Frequency-dependent phase shifts** can be realized - can totally mitigate the beam-split
 - ❑ **Demerits:** a) Each IRS element needs a TTD - **large space & in-efficient hardware** utilization
b) Requires high-resolution TTDs for precise delay control - **requires more power**
 - ❑ TTD-enabled IRS architectures are **not practically very attractive!** 😞
- **Mitigating** beam-split **via distributed approach** - retain phase shifters @ IRS
- **Exploiting** beam-split via an opportunistic OFDMA scheme\$\$

** R. Su, L. Dai, and D. W. K. Ng, "Wideband Precoding for RIS-Aided THz Comm.," in IEEE TCOM, Jun. 2023

\$\$ P. Siddhartha, L. Yashvanth, and Chandra R. Murthy, "Exploiting Beam-split Effects in IRS-aided Systems via OFDMA", accepted for presentation at IEEE ICASSP, Hyderabad, India, Apr. 2025

Mitigating Beam-Split via Distributed IRSs

- Idea: *Parallelize* the *serial* spatial delays of *centralized IRS* via *distributed IRSs*!



- S IRSs with M elements and the same total IRS elements: $N = SM$
- Modified delay spread: $\Delta\tau^D = \max_s \left\{ (M - 1) \frac{d}{c} (\sin(\psi_s) - \sin(\omega_s)) \right\} \approx \frac{\Delta\tau^C}{S}$
- The reduction in SWE also *reduces the beam-split* effect at the UEs

How many elements per IRS?

- The number of elements @ IRS depends on the **tolerable beam split (or beam squint)**

- Channel on the k^{th} sub-carrier (in OFDM):

$$H[k] = \sum_{s=1}^S \sqrt{M} \beta_s \boldsymbol{\theta}_s^H \mathbf{a}_M \left(\sin_{(p)}^{-1} \left\{ \left(1 + \frac{f_k}{f_c} \right) \sin(\phi_s) \right\} \right)$$

- Condition for **ϵ -within beam squint**:

$$|H[0]|^2 = |H[K]|^2 \geq (1 - \epsilon)^2 |H[K/2]|^2$$

Theorem 1: The maximum M for which the channel gain at every IRS is at least $((1 - \epsilon)N)^2$ on every sub-carrier is given by

$$M^* \triangleq \min \left\{ \max \left\{ \left\lfloor \frac{4\sqrt{6\epsilon}}{\pi} \frac{f_c}{W} \right\rfloor, 1 \right\}, N \right\}$$

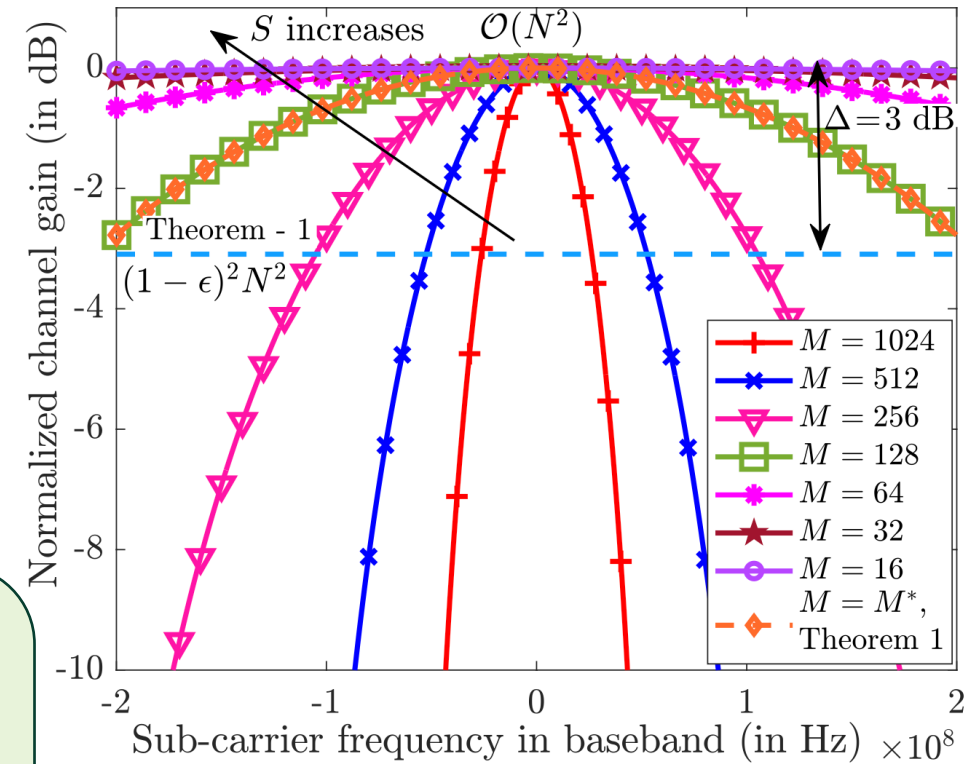


Fig: Theorem - 1 is marked for beam-squint within the HPBW
Setting: $N = 1024$, $M^* = 128$

Spatial vs. Temporal Characteristics of Dist. IRSs

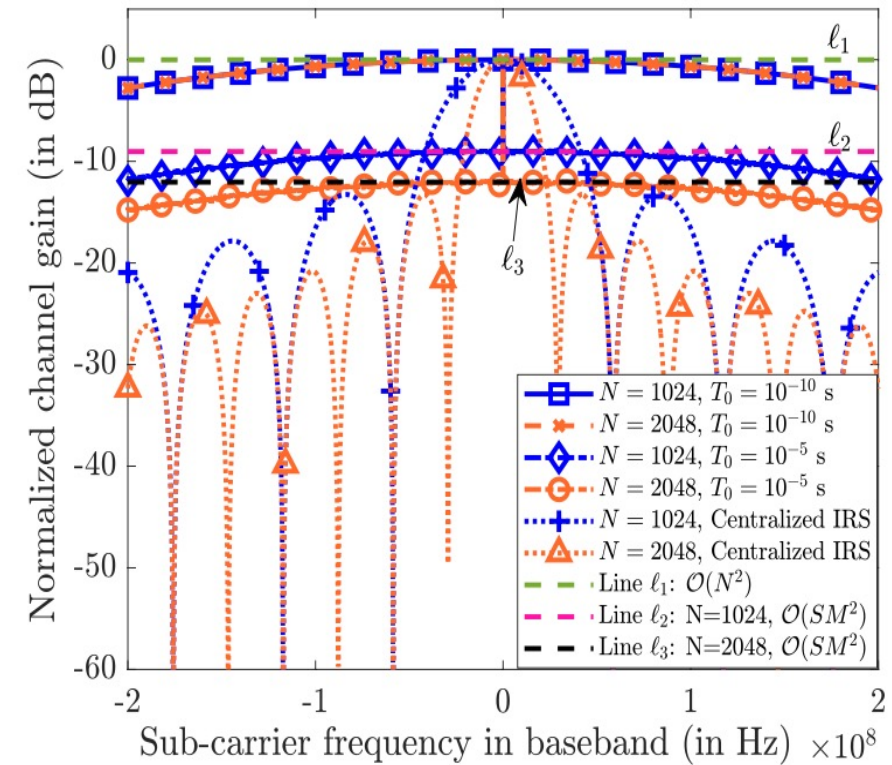
- Multiple IRSs can introduce **temporal delay spread (TDS)** via multiple paths!
- Our solution demonstrates the **interplay** between **spatial** & **temporal** characteristics

Theorem 2: The ergodic SEs of centralized vs. distributed IRS mmWave systems with K SCs:

$$\bar{R}_C \approx \frac{1}{K + N_{CP}^C} \sum_{k=1}^K \log_2 \left(1 + \frac{p_k \sigma_h^2}{\sigma^2} N^2 \text{sinc}^2 \left(N \frac{f_k}{2f_c} \sin(\phi) \right) \right)$$

and

$$\begin{aligned} \bar{R}_D \geq R_{\min} &\triangleq \frac{1}{K + N_{CP}^D} \sum_{k=1}^K \log_2 \left(1 + \frac{p_k \sigma_h^2}{\sigma^2} M^2 (1 - \epsilon)^2 \right. \\ &\quad \times [S^2 \text{sinc}^2(f_k T_0) + S (1 - \text{sinc}^2(f_k T_0))] \Big) \\ &\geq R_{\min}^{\text{L-bound}} \triangleq \frac{1}{K + N_{CP}^D} \sum_{k=1}^K \log_2 \left(1 + \frac{p_k \sigma_h^2}{\sigma^2} S M^2 (1 - \epsilon)^2 \right) \end{aligned}$$



- Distributed IRSs **mitigate B-SP effects even with large TDS** with no deep channel nulls!

How to Position the Multiple IRSs?: A single UE scenario

- Arbitrary IRS locations **reduce the peak again**, although it **mitigates B-SP!**
- Problem: How to determine the **location for the IRSs** so that the **TDS is minimized?**
 - The problem becomes **intractable for a multiple-UE** scenario!
- The **global optimal TDS is zero**, and is achievable as stated in the following result:

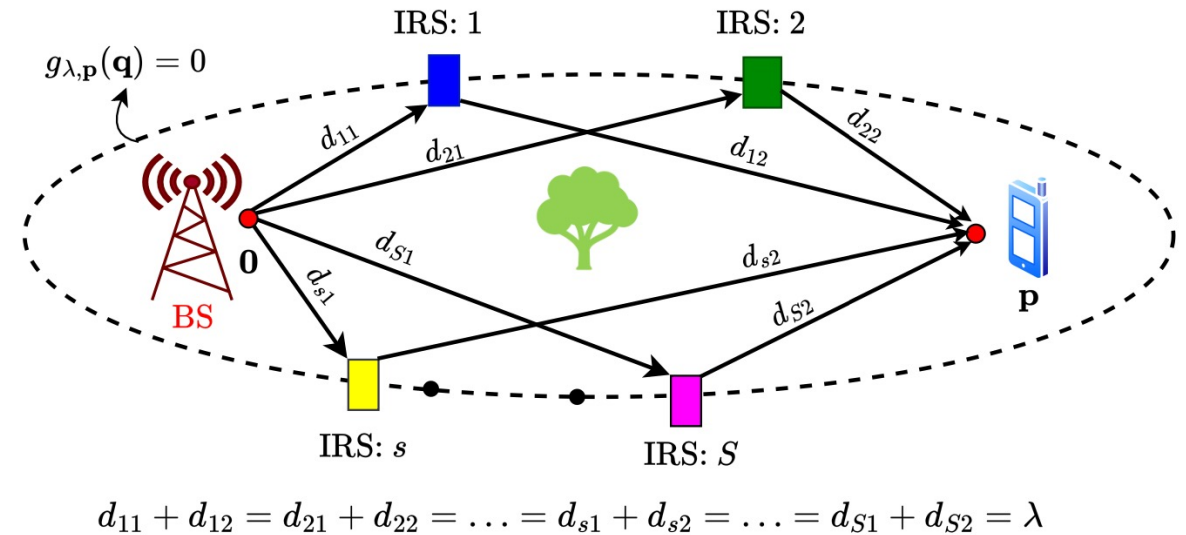
Theorem 3. When BS and UE are located at points $\mathbf{0}$ and $\mathbf{p}$, respectively, the optimal locations of IRSs $(\mathbf{q}_1^*, \dots, \mathbf{q}_S^*)$ as the solution to (P1) satisfy the equation:

$$g_{\lambda, \mathbf{p}}(\mathbf{q}) \triangleq \left\| \mathbf{R}_{\omega, \mathbf{p}}^{\lambda} \left(\mathbf{q} - \frac{\mathbf{p}}{2} \right) \right\|_2^2 - 1 = 0, \quad (35)$$

where we define $\omega \triangleq \tan^{-1} \left(\frac{p_2}{p_1} \right)$, and

$$\mathbf{R}_{\omega, \mathbf{p}}^{\lambda} \triangleq \begin{bmatrix} \frac{2 \cos(\omega)}{2 \sin(\omega)} & \frac{2 \sin(\omega)}{2 \cos(\omega)} \\ \frac{\lambda}{\sqrt{\lambda^2 - \|\mathbf{p}\|_2^2}} & -\frac{\lambda}{\sqrt{\lambda^2 - \|\mathbf{p}\|_2^2}} \end{bmatrix}, \quad (36)$$

with $\lambda \in \mathbb{R}_+$ being a constant satisfying $\lambda > \|\mathbf{p}\|_2$. Further, when the points in $\mathcal{Q}$ satisfy (35), the optimal TDS is $T_d^{D*} = 0$.



Ellipse is the locus of 0-TDS locations for the IRSs

How to Position the Multiple IRSs with multiple UEs?

- For a single UE case, **IRS placement** on the **circumference of an ellipse** yields **0-TDS**
- **A pragmatic choice:** place the IRS on an ellipse with BS and centroid of the area as foci
- Assuming a hotspot region for the UEs,
 - What is the **TDS at a candidate UE** location within the hotspot? And **Achievable SE**?

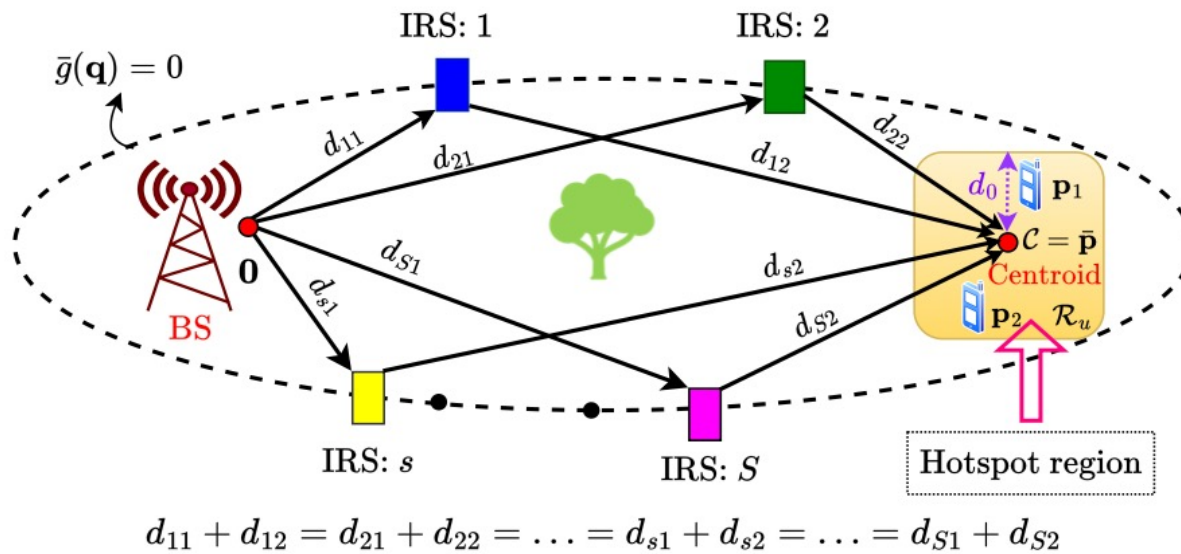


Fig: Geometric locus of IRS locations

Theorem 4: When the UEs are located within a hotspot of radius d_0 , and the IRSs are located on an ellipse as shown in the figure, the avg. sum rate obeys

$$\bar{R}_D^{\text{opt}} \geq R_{\min}^{\text{opt}} \triangleq \frac{1}{K + N_{CP}} \sum_{k=1}^K \log_2 \left(1 + \frac{p_k \sigma_h^2}{\sigma} M^2 (1 - \epsilon)^2 \right. \\ \left. \times \left[S^2 \text{sinc}^2 \left(\frac{2\sqrt{2}d_0 f_k}{c} \right) + S \left\{ 1 - \text{sinc}^2 \left(\frac{2\sqrt{2}d_0 f_k}{c} \right) \right\} \right] \right)$$

Numerical Illustration

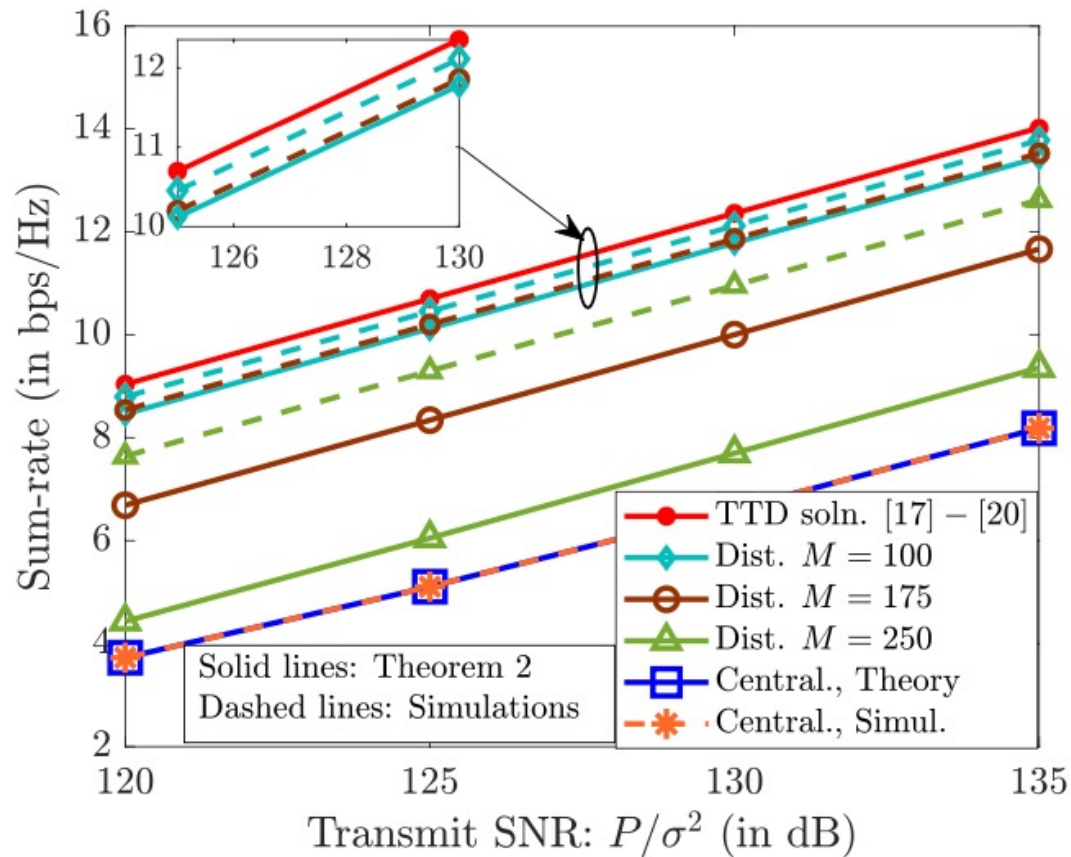


Fig: At a centroid UE, $N = 1024$

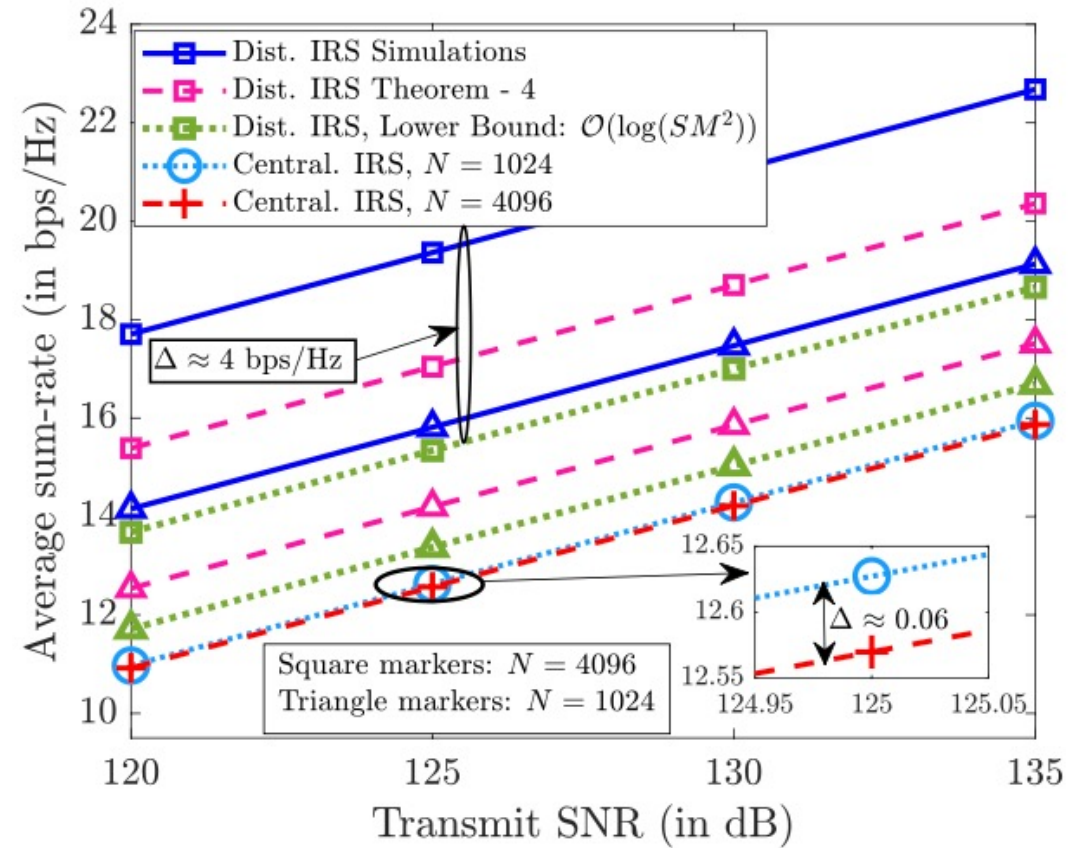


Fig: Avg. across UEs within hotspot

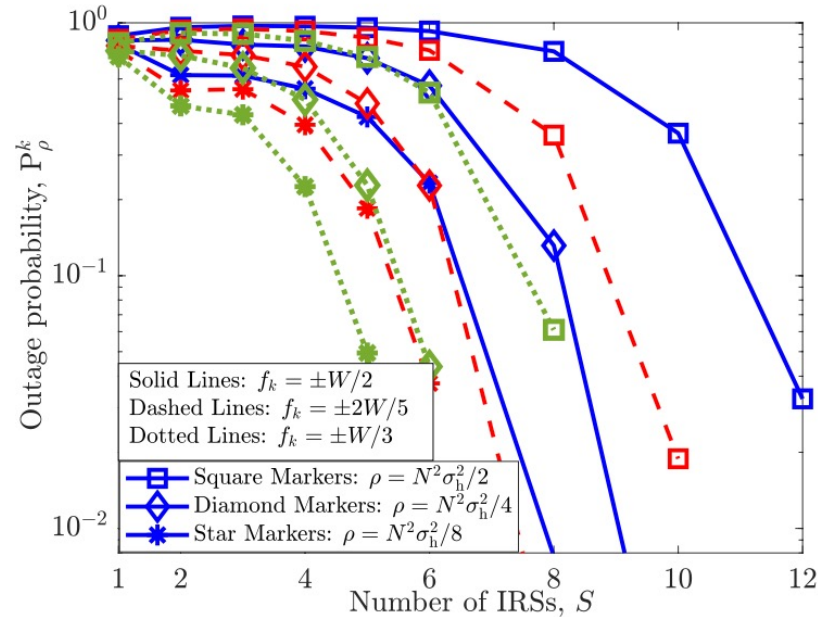
- The rate **increases with # IRSs** and grows as $\mathcal{O}(\log(N^2))$
- Our design performs as well as the TTDs: **low-complex** 😊

➤ **controls SWE/beam-split,**
➤ **achieves full $\mathcal{O}(N^2)$ gain,**
➤ **incurs low-complexity**

Angle Diversity Benefits of Distributed IRSs

- Beam-split induced outage:

$$\mathbb{P}_\rho^k \triangleq \Pr(\{\phi_1, \phi_2, \dots, \phi_S : |H[k]|^2 \leq \rho\})$$



- Outage prob. Exponentially decays in # IRSs - angle diversity
- Bounded away from 0 for centralized IRS
- B-SP happens less likely with more IRSs!

Theorem 5: If the IRSs are tuned to the carrier frequency, the ρ –outage prob. of

a) M-elements S-distributed IRSs obeys

$$\mathbb{P}_\rho^k \leq e^{-2S \left(\left\{ 1 - \frac{\sqrt{\rho}}{SM\zeta\sigma_h^{\min}} \right\} \frac{24f_c^2}{\pi^2 M^2 f_k^2} - \mu_X^{\max} \right)^2},$$

when $\rho \in \left(0, M^2 S^2 \zeta^2 (\sigma_h^{\min})^2 \left(1 - \mu_X^{\max} \frac{\pi^2 M^2 f_k^2}{24f_c^2} \right) \right)$,

$\mu_X^{\max}$ is the max mean angular spread,

and $\sigma_h^{\min}$ is min. path loss across IRSs

b) N-element centralized IRS obeys

$$\mathbb{P}_\rho^k \geq 1 - \frac{1}{\phi_0} \left\{ \sin^{-1} \left(\frac{2f_c \sigma_h}{\pi f_k \sqrt{\rho}} \right) \right\}$$

when $\rho \geq \frac{4f_c^2 \sigma_h^2}{\pi^2 f_k^2 c_0^2}$, c_0 is max. cascaded angle

Summary

- The interplay of a) phased—arrays and b) the spatial wideband effect causes beam-split effects, which degrade the achievable array gain and the throughput.
- Conventional ways of mitigating the beam-split using TTDs are not viable at an IRS
- We propose a distributed approach to mitigate the B-SP effects with phased-array-based IRS
- Idea: control the beam-splitting to tolerable levels by partitioning a large IRS into smaller IRSs
- Spatial delay spread reduces; temporal delay spread increases, which can reduce the flat gain
- Choose the locations of the IRSs appropriately to minimize the temporal delay spread
- In a nutshell, distributed IRSs naturally mitigate beam-split effects at low complexity by
 1. Parallelizing the spatial delays across smaller distributed IRSs
 2. Offering angle diversity benefits to the system

Acknowledgments



Questions?

Thank You!

E-mail(s): yashvanthl@iisc.ac.in